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Radio Frequency Search and Rescue System Using Channel State Information (CSI) Packets

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Background

Search and rescue operations often occur in environments where victims are hidden beneath debris, collapsed structures, or smoke, making detection difficult with traditional methods such as cameras or thermal sensors.

- Wireless signals offer an alternative sensing approach because they can propagate through certain obstacles [1].
- Channel State Information (CSI) from Wi-Fi communication provides detailed measurements of how signals travel between transmitters and receivers [2].
- Human movement alters these signals through reflection and absorption, creating detectable variations [2]. By analyzing CSI using signal processing, it is possible to detect human presence without requiring direct line-of-sight [1], [2].

Fig. 1.Collapsed Structures from Tornadoes [3]

Hypothesis and Research Goals

We hypothesize that variations in Wi-Fi Channel State Information can be analyzed using signal processing to reliably detect human presence and motion in obstructed environments.
The goals of this project are to:

- Achieve ≥80% detection accuracy in controlled indoor environments
- Support detection within a 15–30 meter range
- Maintain response time < 120 seconds (target: < 30 seconds in optimized system)
- Maintain false positive/negative rate < 15%
- Design a system with total cost < $450
- Transmit CSI measurements to a central processing unit for analysis
- Visualize detection results in a real-time heatmap interface
- Demonstrate a portable RF-based sensing system for search and rescue operations

Preliminary Work

- Initial system development focused on establishing reliable CSI data acquisition and communication between sensing nodes and a central processing unit.
- Multiple ESP32 microcontrollers equipped with 2.4 GHz antennas were configured as transmitter and receiver nodes to capture wireless channel measurements. The CSI data collected from these nodes are transmitted to a Raspberry Pi for storage and processing.
- A preliminary signal processing pipeline was implemented to filter noise and extract relevant features from CSI amplitude variations.
- These early experiments demonstrated measurable CSI variations when a person moved within the sensing environment, indicating the feasibility of RF-based human detection.

Methodology

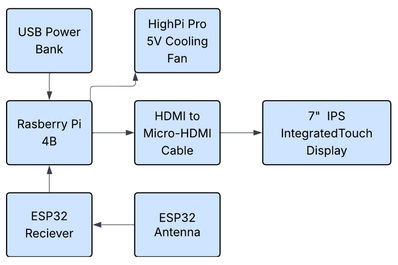

Fig. 2. Central Device Hardware Block Diagram

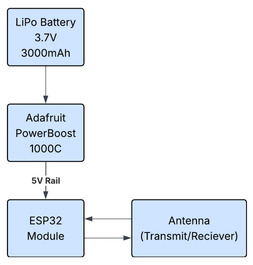

Fig. 3. Node Hardware Block Diagram

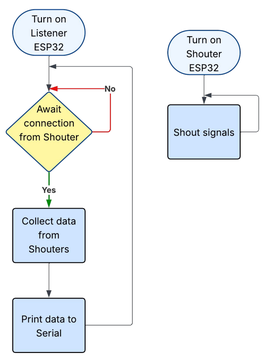

Fig. 4. Software Block Diagram

Hardware Design

1. Multiple ESP32 sensing nodes equipped with Wi-Fi antennas are used to transmit and receive RF signals
2. CSI data from the wireless communication channel is captured at each receiver node
3. A Raspberry Pi serves as the central processing unit to collect CSI
4. An LCD/GUI display is used to present detection results in real time

Signal Processing

1. Raw CSI data is transformed to identify variance over time
2. Raw CSI amplitude data is filtered to remove noise
3. Temporal variance and moving averages are computed
4. Signal disturbances are detected using threshold-based analysis
5. Variance results are visualized as a two-dimensional heatmap showing detected activity within the monitored area.

System Architecture

1. Sensing nodes continuously transmit wireless packets
2. Receiving nodes record CSI measurements
3. Data is streamed to the Raspberry Pi for processing
4. Processed results are displayed through a real-time graphical interface

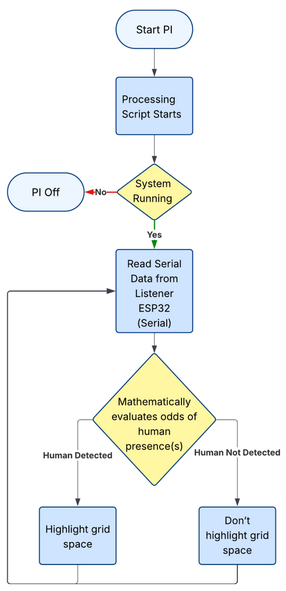

Fig. 5. Software Block Diagram

Results

The prototype system successfully demonstrated the feasibility of CSI-based human detection.
Performance Metrics:

- Detection Accuracy: 80-90% in controlled indoor testing
- Detection Range: Up to 15 meters, target 15-30 meters
- Response Time: 120 seconds, target 30 seconds

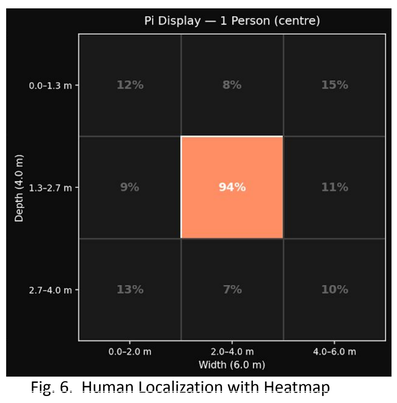

Fig. 6. Human Localization with Heatmap

Key Findings

- Human motion creates detectable disturbances in wireless signals
- Low-cost Wi-Fi hardware can capture these signal variations.
- Signal processing methods alone are sufficient for detecting human presence
- System performance is influenced by environmental noise and multipath effects

The results indicate that wireless CSI sensing can function as a non-contact detection method in obstructed environments.

Discussion

CSI-based sensing provides several advantages for search and rescue applications:

Advantages

- Does not require line-of-sight visibility
- Low-cost, portable, and scalable hardware
- No wearable devices required for detection
- Real-time detection capability

Challenges

- Environmental noise and signal reflections can affect accuracy
- Interference from other wireless devices may impact measurements
- Performance decreases in highly dynamic or cluttered environments

Future work will focus on improving signal processing techniques and expanding the sensing network for better spatial resolution.

Accomplishments and Significance

Accomplishments

- Designed and implemented a multi-node RF sensing system using ESP32 microcontrollers
- Developed a CSI data acquisition pipeline to a Raspberry Pi processing unit
- Implemented signal preprocessing techniques for filtering and variance analysis
- Built a fully-functional prototype capable of detecting human-induced signal disturbances
- Created a real-time visualization interface to display detection results

Significance

- Demonstrates the feasibility of using wireless CSI measurements for human detection in obstructed environments
- Provides a non-contact sensing method that does not require wearable devices or line-of-sight visibility
- Utilizes low-cost (<$450) and widely available Wi-Fi hardware, making the system scalable and portable
- Shows strong potential for search and rescue applications, where victims may be hidden beneath debris or structural barriers
- Establishes a foundation for future improvements such as enhanced localization accuracy, expanded sensing networks, and 3D mapping capabilities

References

[1] C. Uysal and T. Filik, "A new rf sensing framework for human detection through the wall," *IEEE Transactions on Vehicular Technology*, vol. 72, no. 3, pp. 3600–3610, March 2023.
[2] N. Damodaran, E. Haruni, M. Kokhkharova et al., "Device free human activity and fall recognition using wifi channel state information (csi)," *CCF Transactions on Pervasive Computing and Interaction*, vol. 2, p. 1–17, 2020.
[3] "Tornadoes collapse buildings and level homes in Nebraska and Iowa," ABC News (Australia), Apr. 27, 2024. [Online]. Available: https://www.abc.net.au/news/2024-04-27/nebraska-iowa-tornado-destroys-hundreds-of-homes/103776506. [Accessed: Mar. 12, 2026].

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